PYTHON ACTIVITY-BASED LEARNING ASSIGNMENT

ACTIVITY – 1: STUDENT PERFORMANCE ANALYSIS

Case Study No. 17: Student Performance Trend Analysis (Simplified)

1. Problem Statement

Develop a Python program to accept a student's marks in three examinations, calculate the average marks, and identify whether the student's performance is good or needs improvement.

2. Objectives

1. To accept marks in three examinations.
2. To calculate total and average marks.
3. To classify student performance using conditional statements.
4. To display the result in a simple format.

3. Assumptions

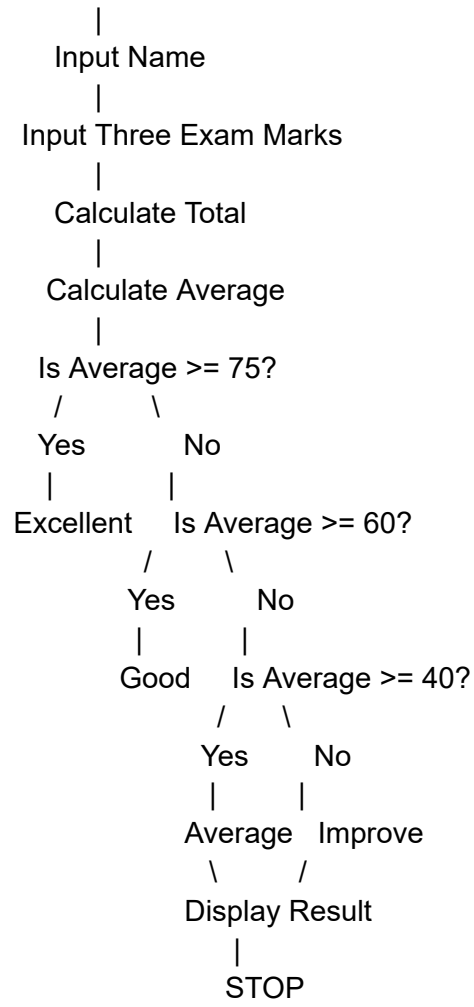
1. Marks are entered out of 100.
2. Marks must be between 0 and 100.
3. An average of 75 or above indicates Excellent performance.
4. An average of 60 or above indicates Good performance.
5. An average of 40 or above indicates Average performance.
6. An average below 40 indicates Needs Improvement.

4. Algorithm

1. Start.
2. Input the student's name.
3. Read marks for three examinations.
4. Check whether all marks are between 0 and 100.
5. Calculate the total marks.
6. Calculate the average marks.
7. Classify performance using if-elif-else statements.
8. Display the name, total, average, and performance.
9. Stop.

5. Flowchart

START



6. Pseudocode

```
BEGIN
   INPUT name
   INPUT mark1, mark2, mark3

   IF any mark < 0 OR any mark > 100
      DISPLAY "Invalid Marks"
   ELSE
      total = mark1 + mark2 + mark3
      average = total / 3

      IF average >= 75
         performance = "Excellent"
      ELSE IF average >= 60
         performance = "Good"
      ELSE IF average >= 40
         performance = "Average"
      ELSE
         performance = "Needs Improvement"
      END IF
```

```
        DISPLAY name
        DISPLAY total
        DISPLAY average
        DISPLAY performance
    END IF
END
```

7. Python Source Code

```
# Activity 1: Student Performance Analysis

name = input("Enter student name: ")

m1 = int(input("Enter marks in Exam 1: "))
m2 = int(input("Enter marks in Exam 2: "))
m3 = int(input("Enter marks in Exam 3: "))

if (m1 < 0 or m1 > 100 or
    m2 < 0 or m2 > 100 or
    m3 < 0 or m3 > 100):

    print("Invalid marks! Enter marks between 0 and 100.")

else:
    total = m1 + m2 + m3
    average = total / 3

    if average >= 75:
        performance = "Excellent"
    elif average >= 60:
        performance = "Good"
    elif average >= 40:
        performance = "Average"
    else:
        performance = "Needs Improvement"

    print("\n--- STUDENT PERFORMANCE REPORT ---")
    print("Student Name:", name)
    print("Total Marks:", total, "/ 300")
    print("Average Marks:", round(average, 2))
    print("Performance:", performance)
```

8. Test Cases

Test Case	Exam 1	Exam 2	Exam 3	Expected Performance

1	90	85	95	Excellent
2	65	70	60	Good
3	45	50	55	Average
4	20	30	25	Needs Improvement
5	110	70	80	Invalid Marks

9. Sample Output

Enter student name: Ravi
Enter marks in Exam 1: 90
Enter marks in Exam 2: 85
Enter marks in Exam 3: 95

--- STUDENT PERFORMANCE REPORT ---
Student Name: Ravi
Total Marks: 270 / 300
Average Marks: 90.0
Performance: Excellent

10. Conclusion

The Student Performance Analysis program was implemented successfully using Python. It calculates total and average marks and classifies student performance using conditional statements. This activity demonstrates the use of variables, operators, input/output statements, and if-elif-else statements.

11. Result

Thus, the Python program for student performance analysis was written and executed successfully.

12. Output Screenshot

Run the program in Python IDLE or VS Code and attach a screenshot of the program and its output.

Input:

```
# Activity 1: Student Performance Analysis

name = input("Enter student name: ")

m1 = int(input("Enter marks in Exam 1: "))
m2 = int(input("Enter marks in Exam 2: "))
m3 = int(input("Enter marks in Exam 3: "))

if (m1 < 0 or m1 > 100 or
    m2 < 0 or m2 > 100 or
    m3 < 0 or m3 > 100):

    print("Invalid marks! Enter marks between 0 and 100.")

else:
    total = m1 + m2 + m3
    average = total / 3

    if average >= 75:
        performance = "Excellent"
    elif average >= 60:
        performance = "Good"
    elif average >= 40:
        performance = "Average"
    else:
        performance = "Needs Improvement"
```

```
else:
    total = m1 + m2 + m3
    average = total / 3

    if average >= 75:
        performance = "Excellent"
    elif average >= 60:
        performance = "Good"
    elif average >= 40:
        performance = "Average"
    else:
        performance = "Needs Improvement"

    print("\n--- STUDENT PERFORMANCE REPORT ---")
    print("Student Name:", name)
    print("Total Marks:", total, "/ 300")
    print("Average Marks:", round(average, 2))
    print("Performance:", performance)
```

Output:

```
Enter student name: arjun
Enter marks in Exam 1: 90
Enter marks in Exam 2: 85
Enter marks in Exam 3: 65

--- STUDENT PERFORMANCE REPORT ---
Student Name: arjun
Total Marks: 240 / 300
Average Marks: 80.0
Performance: Excellent

========== RESTART: C:/Users/Maniratnam/OneDriv
==
Enter student name: siva durga
Enter marks in Exam 1: 95
Enter marks in Exam 2: 85
Enter marks in Exam 3: 89

--- STUDENT PERFORMANCE REPORT ---
Student Name: siva durga
Total Marks: 269 / 300
Average Marks: 89.67
Performance: Excellent
```